

Cyber Security Concepts Hackers and Cyber Crime

Lecture 2



Outline

- The Importance and challenges in Cybersecurity,
- Cyberspace, Cyber threats, Cyberwarfare,
- Cyber Terrorism,
- Cyber Security of Critical Infrastructure,
- Cybersecurity - Organizational Implications.
- Protecting Digital Assets
- Risk management terms, concepts and frameworks
- The Types of Hackers, and Crackers,
- Attack Attributes



The Importance and challenges in Cybersecurity

- Cyber security is important because it protects all categories of data from theft and damage. This includes sensitive data, personally identifiable information (PII), protected health information (PHI), personal information, intellectual property, data, and governmental and industry information systems.

Why is Cyber Awareness Important?

- Cyber crime is a growing trend
- Raise awareness of threats
- As with most crimes the police can't tackle this problem alone
- To encourage reporting
- Promote Government backed scheme 'Cyber Essentials'
- Cyber crime is massively under reported.



Challenges of Cybersecurity

- For effective cyber security, an organization needs to coordinate its efforts throughout its entire information system.
- ***Elements of cyber encompass all of the following:***
 - ❖ **Network security:** The process of protecting the network from unwanted users, attacks and intrusions.
 - ❖ **Application security:** Apps require constant updates and testing to ensure these programs are secure from attacks.
 - ❖ **Endpoint security:** Remote access is a necessary part of business, but can also be a weak point for data. Endpoint security is the process of protecting remote access to a company's network.
 - ❖ **Data security:** Inside of networks and applications is data. Protecting company and customer information is a separate layer of security.
 - ❖ **Identity management:** Essentially, this is a process of understanding the access every individual has in an organization.



Challenges of Cybersecurity

- ❖ **Database and infrastructure security:** Everything in a network involves databases and physical equipment. Protecting these devices is equally important.
- ❖ **Cloud security:** Many files are in digital environments or “the cloud”. Protecting data in a 100% online environment presents a large amount of challenges.
- ❖ **Security for mobile devices:** Cell phones and tablets involve virtually every type of security challenge in and of themselves.
- ❖ **Disaster recovery/business continuity planning:** In the event of a security breach, natural disaster or other event data must be protected and business must go on. For this, you’ll need a plan. End-user education: Users may be employees accessing the network or customers logging on to a company app. Educating good habits (password changes and having a strong password, 2-factor authentication, etc.) is an important part of cybersecurity.



Challenges of Cybersecurity

- ❖ The most difficult challenge in cyber security is the ever-evolving nature of security risks themselves. Traditionally, organizations and the government have focused most of their cyber security resources on perimeter security to protect only their most crucial system components and defend against known threats. Today, this approach is insufficient, as the threats advance and change more quickly than organizations can keep up with. As a result, advisory organizations promote more proactive and adaptive approaches to cyber security.

Cyberspace

- **Cyberspace** is more than the internet, including not only hardware, software, data and information systems, but also people and social interaction within these networks and the whole infrastructure.
- **The International Telecommunication Union (ITU)** uses the term to describe the “systems and services connected either directly to or indirectly to the internet, telecommunications and computer networks.”
- **The International Organisation for Standardisation (ISO)** defining cyber as “the complex environment resulting from the interaction of people, software and services on the internet by means of technology devices and networks connected to it, which does not exist in any physical form.”

Cyber threats,

- ***A cyber or cybersecurity threat*** is a malicious act that seeks to damage data, steal data, or disrupt digital life in general. Cyber threats include computer viruses, data breaches, Denial of Service (DoS) attacks, and other attack vectors.
- **Cyber threats** also refer to the possibility of a successful cyber attack that aims to gain unauthorized access, damage, disrupt, or steal an information technology asset, computer network, intellectual property or any other form of sensitive data. Cyber threats can come from within an organization by trusted users or from remote locations by unknown parties.

Types of Cyber Security Threats

- *Common cyber threats include:*

- ❖ **Malware**

- Malware is software that does malicious tasks on a device or network such as corrupting data or taking control of a system.

- ❖ **Spyware**

- Spyware is a form of malware that hides on a device providing real-time information sharing to its host, enabling them to steal data like bank details and passwords.

- ❖ **Phishing Attacks**

- Phishing is a social-engineering attack that tricks people into giving up credentials, money, or sensitive data (such as personally identifiable information (PII), banking and credit card details and passwords) by impersonating a trusted party (banks, coworkers, vendors, IT).
- Attack vectors include email, SMS (smishing), phone calls (vishing), social media, and fake websites.

- ❖ **Distributed Denial of Service (DDoS) Attacks**

- Distributed denial of service attacks aim to disrupt a computer network by flooding the network with superfluous requests to overload the system and prevent legitimate requests being fulfilled.



Types of Cyber Security Threats

- *Common cyber threats include:*

- ❖ Ransomware

- [Ransomware](#) is a type of malware that denies access to a computer system or data until a ransom is paid.

- ❖ Zero-Day Exploits

- A zero-day exploit is a flaw in software, hardware or firmware that is unknown to the party or parties responsible for patching the flaw.

- ❖ Advanced Persistent Threats

- An advanced persistent threat is when an unauthorized user gains access to a system or network and remains there without being detected for an extended period of time.

- ❖ Trojans

- A trojan creates a backdoor in your system, allowing the attacker to gain control of your computer or access confidential information.



Types of Cyber Security Threats

- *Common cyber threats include:*

- ❖ **Wiper Attacks**

- A wiper attack is a form of malware whose intention is to wipe the hard drive of the computer it infects.

- ❖ **Intellectual Property Theft**

- Intellectual property theft is stealing or using someone else's intellectual property without permission.

- ❖ **Theft of Money**

- Cyber attacks may gain access to credit card numbers or bank accounts to steal money.

- ❖ **Data Manipulation**

- Data manipulation is a form of cyber attack that doesn't steal data but aims to change the data to make it harder for an organization to operate.

- ❖ **Data Destruction**

- Data destruction is when a cyber attacker attempts to delete data.



Types of Cyber Security Threats

- **Common cyber threats include:**

- ❖ **Man-in-the-Middle Attack (MITM Attack)**

- A MITM attack is when an attacker relays and possibly alters the communication between two parties who believe they are communicating with each other.

- ❖ **Drive-by Downloads**

- A drive-by download attack is a download that happens without a person's knowledge often installing a computer virus, spyware, or malware.

- ❖ **Malvertising**

- Malvertising is the use of online advertising to spread malware.

- ❖ **Rogue Software**

- Rogue software is malware that is disguised as real software.

- ❖ **Unpatched Software**

- Unpatched software is software that has a known security weakness that has been fixed in a later release but not yet updated.



Types of Cyber Security Threats

- ***Common cyber threats include:***

- ❖ **Data Centre Disrupted by Natural Disaster**

- The data centre your software is housed on could be disrupted by a natural disaster like flooding.

- ❖ **SQL Injection**

- A Structured Query Language (SQL) injection is a type of cyber attack that results from inserting malicious code into a server that uses SQL. When infected, the server releases information. Submitting the malicious code can be as simple as entering it into a vulnerable website search box.

- ❖ **Password Attacks**

- With the right password, a cyber attacker has access to a wealth of information. Social engineering is a type of password attack that Data Insider defines as “a strategy cyber attackers use that relies heavily on human interaction and often involves tricking people into breaking standard security practices.” Other types of password attacks include accessing a password database or outright guessing.

Cyber Warfare

- **Cyber warfare** is usually defined as a cyber attack or series of attacks that target a country. It has the potential to wreak havoc on government and civilian infrastructure and disrupt critical systems, resulting in damage to the state and even loss of life.
- **Here are some of the main types of cyber warfare attacks**



Espionage



Sabotage



Denial-of-service
(DoS) Attacks



Electrical
Power Grid



Propaganda
Attacks



Economic
Disruption



Surprise
Attacks

Cyber Warfare

- **Here are some of the main types of cyber warfare attacks**

- **Espionage**

Refers to monitoring other countries to steal secrets. In cyber warfare, this can involve using botnets or spear phishing attacks to compromise sensitive computer systems before exfiltrating sensitive information.

- **Sabotage**

Government organizations must determine sensitive information and the risks if it is compromised. Hostile governments or terrorists may steal information, destroy it, or leverage insider threats such as dissatisfied or careless employees, or government employees with affiliation to the attacking country.

- **Denial-of-service (DoS) Attacks**

DoS attacks prevent legitimate users from accessing a website by flooding it with fake requests and forcing the website to handle these requests. This type of attack can be used to disrupt critical operations and systems and block access to sensitive websites by civilians, military and security personnel, or research bodies.

Cyber Warfare

- **Here are some of the main types of cyber warfare attacks**

- **Electrical Power Grid**

Attacking the power grid allows attackers to disable critical systems, disrupt infrastructure, and potentially result in bodily harm. Attacks on the power grid can also disrupt communications and render services such as text messages and communications unusable.

- **Propaganda Attacks**

Attempts to control the minds and thoughts of people living in or fighting for a target country. Propaganda can be used to expose embarrassing truths, spread lies to make people lose trust in their country, or side with their enemies.

Cyber Warfare

- **Here are some of the main types of cyber warfare attacks**

- **Economic Disruption**

Most modern economic systems operate using computers. Attackers can target computer networks of economic establishments such as stock markets, payment systems, and banks to steal money or block people from accessing the funds they need.

- **Surprise Attacks**

These are the cyber equivalent of attacks like Pearl Harbor and 9/11. The point is to carry out a massive attack that the enemy isn't expecting, enabling the attacker to weaken their defenses. This can be done to prepare the ground for a physical attack in the context of hybrid warfare.

Cyber Terrorism

- ❖ **Cyber terrorism** is often defined as any premeditated, politically motivated attack against information systems, programs and data that threatens violence or results in violence. The definition is sometimes expanded to include any cyber attack that intimidates or generates fear in the target population.

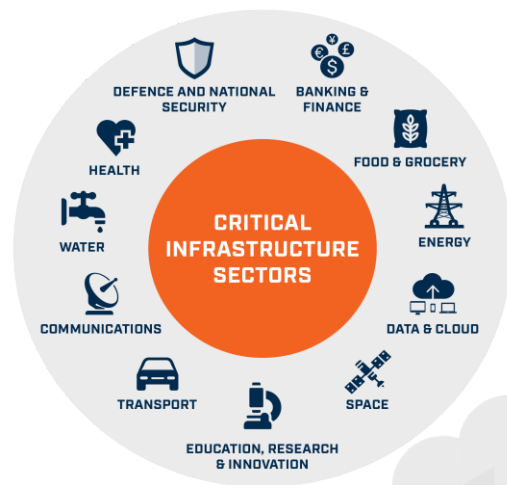
- ❖ The nature of cyberterrorism covers conduct involving computer or Internet technology that:
 - 1) is motivated by a political, religious or ideological cause
 - 2) is intended to intimidate a government or a section of the public to varying degrees
 - 3) seriously interferes with infrastructure

Cyber Security of Critical Infrastructure

- Critical infrastructure describes the physical and cyber systems and assets that are so vital to the countries that their incapacity or destruction would have a debilitating impact on physical or economic security or public health or safety.

Critical infrastructure sector

Chemical	Dams	Financial services	Information technology
Commercial facilities	Defense industrial base	Food and agriculture	Nuclear reactors, nuclear materials, and waste
Communications	Emergency services	Government facilities	Transportation systems
Critical manufacturing	Energy	Health care, public health	Water and wastewater systems



Cybersecurity - Organizational Implications

- In the global environment with continuous network connectivity, the possibilities for cyberattacks can emanate from sources that are local, remote, domestic or foreign. They could be launched by an individual or a group. They could be casual probes from hackers using personal computers (PCs) in their homes, hand-held devices, or intense scans from criminal groups.

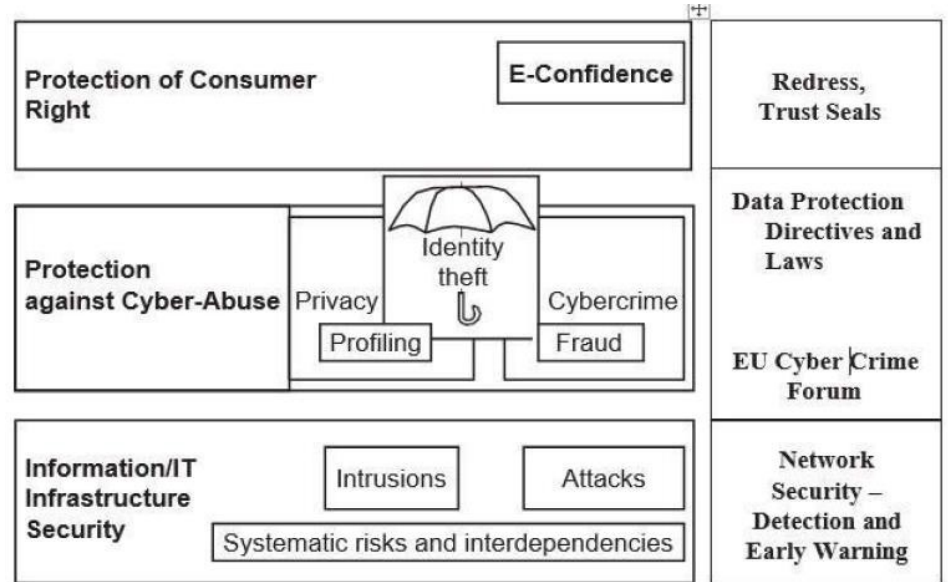


Fig: A cyber security perspective. EU is the European Union.

The Cost of Cybercrimes

- Cybercrimes cost a lot to organizations.
- When a cybercrime incidence occurs, there are a number of internal costs associated
- with it for organizations and there are organizational impacts as well.

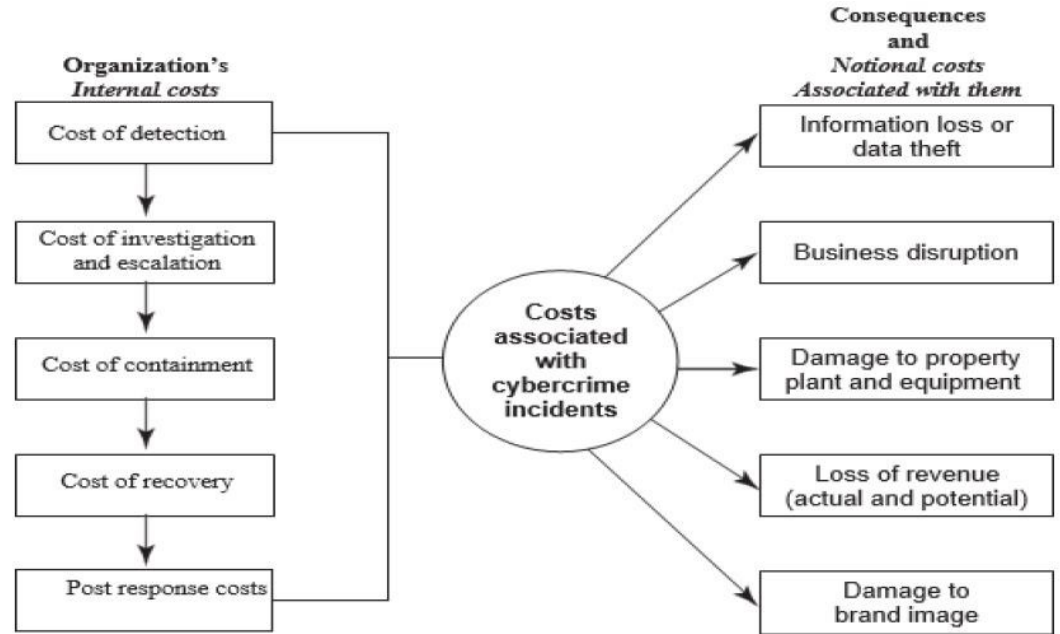


Fig: Cost of cybercrimes.

Protecting Digital Assets

- In **cybersecurity** frameworks, both the National Institute of Standards and Technology (NIST) and the European Union Agency for Network and Information Security (ENISA) have identified five key functions necessary for the protection of digital assets.
- These functions coincide with incident management methodologies and include the following activities:
- **Identify**: Use organizational understanding to minimize risk to systems, assets, data and capabilities.
- **protect**: Design safeguards to limit the impact of potential events on critical services and infrastructure.
- **Detect**: Implement activities to identify the occurrence of a cybersecurity event.
- **Respond**: Take appropriate action after learning of a security event.
- **Recover**: Plan for resilience and the timely repair of compromised capabilities and services.

Risk management terms, concepts and frameworks

➤ Risk

- The core duty of **cybersecurity** is to identify, mitigate and manage **cyberrisk** to an organization's digital assets.:
- **Assessing risk** is one of the most critical functions of a cybersecurity organization.
- Effective policies, security implementations, resource allocation and incident response preparedness are all dependent on → understanding the risk and threats an organization faces.
- Using a **risk-based approach** to cybersecurity allows more informed decision-making to protect the organization and to apply limited budgets and resources effectively.



Risk management terms, concepts and frameworks

Approaches to cybersecurity

Generally, there are three different approaches to implementing cybersecurity:

- 1) **Compliance-based**—Also known as standards-based security, this approach relies on regulations or standards to determine security implementations. Controls are implemented regardless of their applicability or necessity, which often leads to a “checklist” attitude toward security.
- 2) **Risk-based**—Risk-based security relies on identifying the unique risk a particular organization faces and designing and implementing security controls to address that risk above and beyond the entity’s risk tolerance and business needs.
- 3) **Ad hoc**—An **ad hoc** approach simply implements security with no particular rationale or criteria. Ad hoc implementations may be driven by vendor marketing, or they may reflect insufficient subject matter expertise, knowledge or training when designing and implementing safeguards.



Risk management terms, concepts and frameworks

➤ key terms and definitions

- *There are many potential definitions of risk—some general and others more technical. Additionally, it is important to distinguish between a risk and a threat. Although many people use the words threat and risk synonymously, they have two very different meanings:*

Risk—The combination of the probability of an event and its consequence (International Organization for Standardization/International Electrotechnical Commission [ISO/IEC] 73).

Risk is mitigated through the use of controls or safeguards.

Different industries and professions have adopted various tactics based upon the following criteria:

- ❖ Risk tolerance
- ❖ Size and scope of the environment in question
- ❖ Amount of data available

Risk management terms, concepts and frameworks

key terms and definitions

Threat—Anything (e.g., object, substance, human) that is capable of acting against an asset in a manner that can result in harm.

Asset—Something of either tangible or intangible value that is worth protecting, including people, information, infrastructure, finances and reputation.

Vulnerability—A weakness in the design, implementation, operation or internal control

of a process that could expose the system to adverse threats from threat events

Residual risk—Even after safeguards are in place, there will always be residual risk, defined as the remaining risk after management has implemented a risk response.

Inherent risk—The risk level or exposure without taking into account the actions that management has taken or might take (e.g., implementing controls)

Risk management terms, concepts and frameworks

➤ Third-party Risk

- ✓ *Cybersecurity can be more difficult to control when third parties are involved, especially when different entities have different security cultures and risk tolerances.*
- ✓ *No organization exists in a vacuum, and information must be shared with other individuals or organizations, often referred to as third parties.*
- ✓ *It is important to understand third-party risk, such as information sharing and network access, as it relates to cybersecurity.*

The Hackers, and Crackers – Definition

Hacking:

Hacking may be defined as the technique or planning which is done to get an access to unauthorized systems. Simply we can say to gaining access to a network or a computer for illegal purposes. The person who does that are very intelligent and skilled in computers.

The person who are skilled in Hacking are divided into 2 categories:

1.Hackers

- Hackers are kind of good peoples who does hacking for the good purpose and to obtain more knowledge from it.
- They generally find loop holes in the system and help them to cover the loop holes.
- Hackers are generally programmers who obtain advance knowledge about operating systems and programming languages.
- These people never damage or harm any kind of data.

The Hackers, and Crackers – Definition

2. Crackers

- Crackers are kind of bad people who breaks or violates the system or a computer remotely with bad intentions to harm the data and steal it.
- Crackers destroy data by gaining an unauthorized access to the network.
- Their works are always hidden as they are doing illegal stuff.
- Bypasses passwords of computers and social media websites, can steal your bank details and transfer money from bank.

Difference between Hackers and Crackers :

Hacker

The good people who hack for knowledge purposes.

They are skilled and have an advance knowledge of computers OS and programming languages.

They work in an organisation to help protecting there data and giving them expertise on internet security.

Hackers share the knowledge and never damages the data.

Hackers are the ethical professionals.

Hackers program or hacks to check the integrity and vulnerability strength of a network.

Hackers have legal certificates with them e.g CEH certificates.

Cracker

The evil person who breaks into a system for benefits.

They may or may not be skilled, some of crackers just knows a few tricks to steal data.

These are the person from which hackers protect organisations .

If they found any loop hole they just delete the data or damages the data.

Crackers are unethical and want to benefit themselves from illegal tasks.

Crackers do not make new tools but use someone else tools for there cause and harm the network.

Crackers may or may not have certificates, as there motive is to stay annonymous.

The Types of Hackers, and Crackers

Threat Agents:

- Corporations
- Nation States
- Hacktivists
- Cyberterrorists
- Cybercriminals
- Script kiddies
- Online Social hackers

The Types of Hackers, and Crackers

❖ Threat Agents:

➤ **Corporations:-**

- Corporations have been known to breach security boundaries and perform malicious acts to gain a competitive advantage.

➤ **Nation States:-**

- Nation states often target government and private entities with a high level of sophistication to obtain intelligence or carry out other destructive activities.

The Types of Hackers, and Crackers

❖ Threat Agents:

➤ **Hacktivists:-**

- Although they often act independently, politically motivated hackers may target specific individuals or organizations to achieve various ideological ends.

➤ **Cyberterrorists:-**

- Characterized by their willingness to use violence to achieve their goals, cyberterrorists frequently target critical infrastructures and government groups.

The Types of Hackers, and Crackers

❖ Threat Agents:

➤ **Cybercriminals:-**

- Motivated by the desire for profit, these individuals are involved in fraudulent financial transactions.

➤ **Cyberwarriors:-**

- Often likened to hacktivists, cyberwarriors, also referred to as cyberfighters, are nationally motivated citizens who may act on behalf of a political party or against another political party that threatens them.

The Types of Hackers, and Crackers

❖ Threat Agents:

➤ **Script kiddies:-**

- Script kiddies are young individuals who are learning to hack; they may work alone or with others and are primarily involved in code injections and distributed denial-of-service (DDoS) attacks.

➤ **Online Social hackers:-**

- Skilled in social engineering, these attackers are frequently involved in cyberbullying, identity theft and collection of other confidential information or credentials.

The Types of Hackers, and Crackers

❖ *Hackers can be classified into different categories such as **white hat, black hat, and grey hat**, based on their intent of hacking a system.*

- **1) White Hat Hackers**

- White Hat hackers are also known as **Ethical Hackers**.
- They never intent to harm a system, rather they try to find out weaknesses in a computer or a network system as a part of penetration testing and vulnerability assessments.
- Ethical hacking is not illegal and it is one of the demanding jobs available in the IT industry.
- There are numerous companies that hire ethical hackers for penetration testing and vulnerability assessments.

The Types of Hackers, and Crackers

2) Black Hat Hackers

- Black Hat hackers, also known as **crackers**, are those who hack in order to gain unauthorized access to a system and harm its operations or steal sensitive information.
- Black Hat hacking is always illegal because of its bad intent which includes:
 - stealing corporate data,
 - violating privacy,
 - damaging the system,
 - blocking network communication, etc.

3) Grey Hat Hackers

- Grey hat hackers are a blend of both black hat and white hat hackers.
- They act without malicious intent but for their fun,
- They exploit a security weakness in a computer system or network without the owner's permission or knowledge.
- Their intent is to bring the weakness to the attention of the owners and getting appreciation or a little bounty from the owners.

The Types of Hackers, and Crackers

4) Red Hat Hackers

- Red hat hackers are again a blend of both black hat and white hat hackers.
- They are usually on the level of hacking government agencies, top- secret information hubs, and generally anything that falls under the category of sensitive information.

5) Blue Hat Hackers

- A blue hat hacker is someone outside computer security consulting firms who is used to bug-test a system prior to its launch.
- They look for loopholes that can be exploited and try to close these gaps.
- Microsoft also uses the term **BlueHat** to represent a series of security briefing events.

6) Script Kiddie

- A script kiddie is a non-expert who breaks into computer systems by using pre-packaged automated tools written by others,
- Usually with little understanding of the underlying concept, hence the term **Kiddie**.

7) Hacktivist

- A hacktivist is a hacker who utilizes technology to announce a social, ideological, religious, or political message.
- In general, most hacktivism involves website defacement or denial-of-service attacks.

Self Assignment

Case Study:

Stuxnet attack components

Internet of Things (IoT) Attacks

Thank you!

